### 1.2.3 ▶ ▶ ▶ ▶ Centripetal Acceleration

When an object moves in a circle at a constant velocity, it experiences an inward acceleration due the direction of the velocity vector changing with time. In this section, we derive this acceleration known as the centripetal acceleration.

First, consider a particle moving in a circle from a point $A$ to a point $B$.

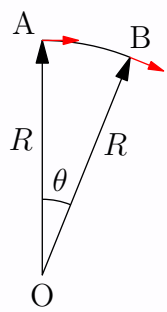


**Fig. 1.1**

Where the red vectors are the velocity vectors of the particle at points A and B. Connecting the tails of these two vectors together, we can find the change in velocity and thus the acceleration of this particle.

**(a)** Figure a **(b)** Figure b

**Fig. 1.2**

For small angles $\theta$, Fig.1.2b simply becomes a triangle[4], giving us a pair of similar triangles. Thus, we have

$$\frac{v}{\Delta v} = \frac{R}{S}. \tag{1.29}$$

Using $S = v\Delta t$, we have

$$\frac{v}{\Delta v} = \frac{R}{v\Delta t}$$

$$\frac{\Delta v}{\Delta t} = \frac{v^2}{R}$$

$$\boxed{a_c = \frac{v^2}{R}}. \tag{1.30}$$

Which is the magnitude of the centripetal acceleration of an object moving in a circle ★

[4]See A.2.1 for a more rigorous proof